

The Emergence of Cooperation

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Recap

In Axelrod's round-robin tournaments, the winning strategy was **Tit for Tat** (TFT): cooperate on move one, then copy your opponent's last move.

Axelrod identified four properties that successful strategies share:

- **Nice:** never be the first to defect
- **Retaliatory:** respond to defection quickly
- **Forgiving:** return to cooperation after punishing
- **Clear:** be legible to your opponent

But the tournament just ranks strategies against a fixed field.

Here's a question that might be more important in realistic situations:

Which strategies *stay* successful over time?

Imagine this variant of the tournament.

Start not with 1 copy of each strategy but 100.

After each round, strategies that did well get **more copies** in the next round, and strategies that did poorly get **fewer copies**.

This is an *ecological* simulation: the population of strategies changes based on fitness.

What Difference Does it Make?

What distinctive factors do **you** have to consider when thinking about what will do well in this version of the game?

Two things (at least) pop out:

1. Performance against conspecifics matters.
2. Scoring big against easybeats does not matter.

How might **you** try to design a strategy to exploit this?

Two things jump out about the type of 'hunter'/'detector' strategies that work around here.

1. 'Handshake' strategies that script a few moves then cooperate with conspecifics and defect against rivals might have advantages.
2. There is little benefit to detecting if someone is particularly vulnerable to exploitation; after a while they'll have been exploited.

Axelrod ran exactly this simulation.

The question: does TFT dominate, or does the changing environment eventually undermine it?

Another question that's come up separately: should we have each strategy interact with **all** other strategies or just with their **neighbors**?

We'll mostly talk about the 'interact with everyone' version here, but the 'neighbors' version might be more useful for modelling some situations.

The Evolutionary Setting

Axelrod and Hamilton (1981) asked: can cooperation evolve by **natural selection**, with no central authority, no communication, and no altruism?

The setup: think of strategies not as choices but as **heritable traits**. Organisms that do better reproduce more. Strategies that earn more points spread in the population.

The Prisoner's Dilemma becomes a model of biological interaction: two organisms cooperate (e.g., by helping each other) or defect (by freeloading).

Payoffs are in **fitness**, i.e., reproductive success, not dollars.

Why do **you** think would any organism ever cooperate?

Classical evolutionary theory had two answers:

- **Kin selection:** you help relatives because they share your genes
(Hamilton's rule: $rb > c$)
- **Group selection:** groups with cooperators outcompete groups of defectors

Axelrod and Hamilton proposed a third mechanism: **reciprocal altruism** arising from repeated interaction, even among unrelated individuals.

For cooperation to be stable, the **shadow of the future** must be long enough.

More precisely: let w be the probability that two organisms will interact again. Cooperation is stable if w is large enough that the future gains from mutual cooperation outweigh the short-term gains from defection.

$$w > \frac{T - R}{T - P}$$

Where T = temptation payoff, R = reward for mutual cooperation, P = punishment for mutual defection.

When interactions are likely to continue, cooperation can sustain itself.

It's impossible to work out what might work best against arbitrary strategies; there are too many alternatives.

So a lot of focus has been on the possibility of **invasion**.

What happens if a community is homogenous, then a small number of a new strategy appears?

Will this new strategy survive, or even take over?

Start with a population of **All Defectors**. Can a small cluster of TFT players invade?

Answer: **yes**, if there are enough TFT players and they tend to interact with each other.

When TFT meets TFT: mutual cooperation earns R per round for many rounds. When TFT meets All D: TFT defects back and both earn P .

If TFT players cluster together, geographically, socially, by kin, they can earn enough against each other to outperform the defectors, even accounting for losses at the boundary.

This clustering is what Axelrod and Hamilton call the key to invasion.

- A **single** TFT mutant in an All D population does badly: it cooperates once, gets exploited, then defects—but there's no one to cooperate with afterward.
- A **small cluster** of TFT mutants does well *against each other*, and that advantage can outweigh the losses at the boundary with defectors.
- As TFT spreads, the advantage grows: there are more TFT players to cooperate with.

Can TFT Resist Invasion?

Once TFT is established, can a mutant defector strategy invade the TFT population?

No; well, at least not easily.

A rare defector in a TFT population gets one round of exploitation, then faces retaliation for every subsequent round. The gains from that first defection are outweighed by the costs of the long punishment period.

TFT is **evolutionarily stable** (in the informal sense): it resists invasion by defectors once it has taken hold.

That said, if there are enough invaders, and they behave differently towards in group and out group members, they may have a chance. But there need to be a *lot* of invaders.

In the ecological version of Axelrod's tournament:

- Initially, all 63 strategies have equal representation.
- Each “generation,” strategies are reweighted by their total score.
- Nice strategies (which cooperated well with each other) **grew in population share**.
- Nasty strategies (which exploited others early) **shrank**, because as nice strategies became rarer, there were fewer victims to exploit.

The final population was dominated by nice strategies, with TFT among the most successful.

TFT is not universally dominant. Axelrod's result depends on:

- The *composition* of the field (what other strategies are present);
- The payoff structure (how large T and P are relative to R); and
- The length and continuity of interactions.

In some environments, more generous variants of TFT (e.g., **Generous TFT**) outperform standard TFT.

The deeper lesson is not “TFT always wins”.

It's that **niceness, retaliation, and forgiveness** are the robust ingredients.

Applications

Axelrod's evolutionary framework has been applied across a remarkable range of fields:

- **Biology:** mutualism between species (cleaner fish, fig wasps, mycorrhizal fungi);
- **International relations:** arms control treaties, trade agreements;
- **Business:** repeated commercial dealings and reputation; and
- **Everyday social life:** trust, reciprocity, and social norms.

Axelrod and Hamilton pointed to **cleaner fish** as a real-world case.

Cleaner wrasse remove parasites from “client” fish, who could easily eat them.

But they don't get eaten, and so clients return repeatedly to the same cleaner.

This is iterated PD in the wild: each interaction is potentially exploitable, but repeated interaction makes cooperation stable.

Similar patterns appear in fig trees and fig wasps, and in mycorrhizal fungi that trade nutrients with plant roots.

Two superpowers in an arms race arguably face a PD structure: both would prefer mutual restraint, but each has an incentive to build while the other restrains.

The iterated setting makes cooperation possible: arms *control* agreements work as TFT-like strategies: comply as long as the other side complies, retaliate (verifiably) if they cheat.

Axelrod used this to explain how **Live and Let Live** emerged on the Western Front in WWI.

Small units facing each other across the trenches cooperated locally, even without orders to do so.

Here we have to be careful about making sure we are really dealing with PD games.

Often, what people identify as PD is really a **Stag Hunt**.

Think about the practice of walking on right in the halls (and especially staircases) around here. That's cooperation, but not a PD.

Repeated commercial relationships support cooperation that one-shot interactions can't.

- A supplier who cheats once loses a long-term customer, so the shadow of the future disciplines behavior.
- Online platforms (eBay, Airbnb) create iterated PD conditions artificially, through **reputation systems** that make interaction history visible. (See *The Oridinal Society* for the downstream effects of this.)
- Professional networks and trade associations do something similar, though without the ordinal aspect.

The evolutionary story also illuminates how **norms** function.

A norm against defecting (i.e., cheating) is a way of making retaliation credible. It also motivates **altruistic punishment**, which strongly enforces norms.

Axelrod's framework suggests that norms didn't need to be designed from the top down. They could have **evolved** from repeated local interactions, spreading as cooperating communities outperformed defecting ones.

Variations and Complications

What If the Game Is Noisy?

Axelrod's original setup assumed **perfect implementation**: if you choose to cooperate, your opponent definitely sees cooperation.

But real interactions are often noisy. This is especially for interactions between groups.

Noisy IPD: players submit their moves, but each move is implemented incorrectly with some small probability ε (e.g., $\varepsilon = 0.01$).

This small change **breaks TFT badly**.

Suppose both players are playing TFT, and a noise event causes player A's cooperation to be received as defection.

Now player B retaliates (correctly, from their perspective). A then retaliates B's retaliation. B retaliates again. . .

A single noise event causes an **infinite defection spiral** even though neither player actually chose to defect.

The very property that makes TFT robust in the clean game, immediate retaliation, becomes catastrophic when moves are sometimes garbled.

One fix: **Generous TFT** (GTFT). Instead of always retaliating after a defection, GTFT occasionally forgives: after a defection, it cooperates with probability $p > 0$.

This breaks the death spiral: a noise event triggers retaliation, but GTFT eventually cooperates again, and mutual cooperation is restored.

GTFT outperforms TFT in noisy environments, and remains stable against invasion by defectors as long as p is not too large.

Another fix: **Contrite TFT**. After a noise event causes you to defect accidentally, you “apologize” by cooperating even if your opponent retaliates.

The key thing here is that each player knows what each player **played**, and the players each know what they intended to play.

So a player can react to their own ‘mistakes’.

If both players are Contrite TFT:

- Noise event: A accidentally defects;
- B retaliates (correctly);
- A recognizes it was an accident, and cooperates being *contrite*;
- Finally, B sees A cooperate, resumes cooperation.

The death spiral never starts, because one player backs down rather than escalating.

Nowak and Sigmund (1993) proposed a different strategy that outperforms TFT in noisy settings: **Pavlov** (or Win-Stay, Lose-Shift).

The rule: repeat your last move if you got a good payoff (i.e., 3 or more), switch if you got a bad payoff.

- If you both cooperated (R): cooperate again, since you got 3.
- If you defected and they cooperated (T): defect again, since you got 5.
- If you both defected (P): cooperate; getting 1 leads to changes.
- If you cooperated and they defected (S): defect, since getting 0 really leads to shifting.

Pavlov is **not nice**: it will exploit an Always Cooperate player (cooperate, randomly defect, defect, defect...).

TFT can't expel an Always Cooperating invader. So a community of TFT's with random introductions will eventually include some All C's, and Pavlov will successfully invade it.

Pavlov is also **more robust to noise** than TFT: after an accidental defection, Pavlov's "lose-shift" rule brings it back to cooperation faster.

These variations point toward a general lesson:

There is **no single dominant strategy** for the iterated PD across all environments. The best strategy depends on:

- The noise level in the interaction;
- The composition of the population (who else is playing?);
- The discount factor (how much the future matters); and
- The payoff structure

What *does* seem robust: some combination of cooperation, retaliation, and forgiveness. The exact balance shifts with context.

If time permits, consider:

- **Does TFT work in multi-player games?** The two-player structure is crucial. With many players, identifying *who* defected—and retaliating against them specifically—becomes much harder.
- **What about different network structures?** If players only interact with neighbors, spatial structure changes which strategies survive.
- **How does optional participation affect things?** If players can exit the game, it disrupts stable exploitation—and may help cooperation get started.

We'll look at one of the famous use of game-theory to model real-world situations.

- On Monday, we'll look at George Akerlof's use of signalling games to understand the used car market.
- On Wednesday, we'll look at Michael Spence's similar use of signalling games to understand the college wage premium.